Chapter 3

Data and Signals

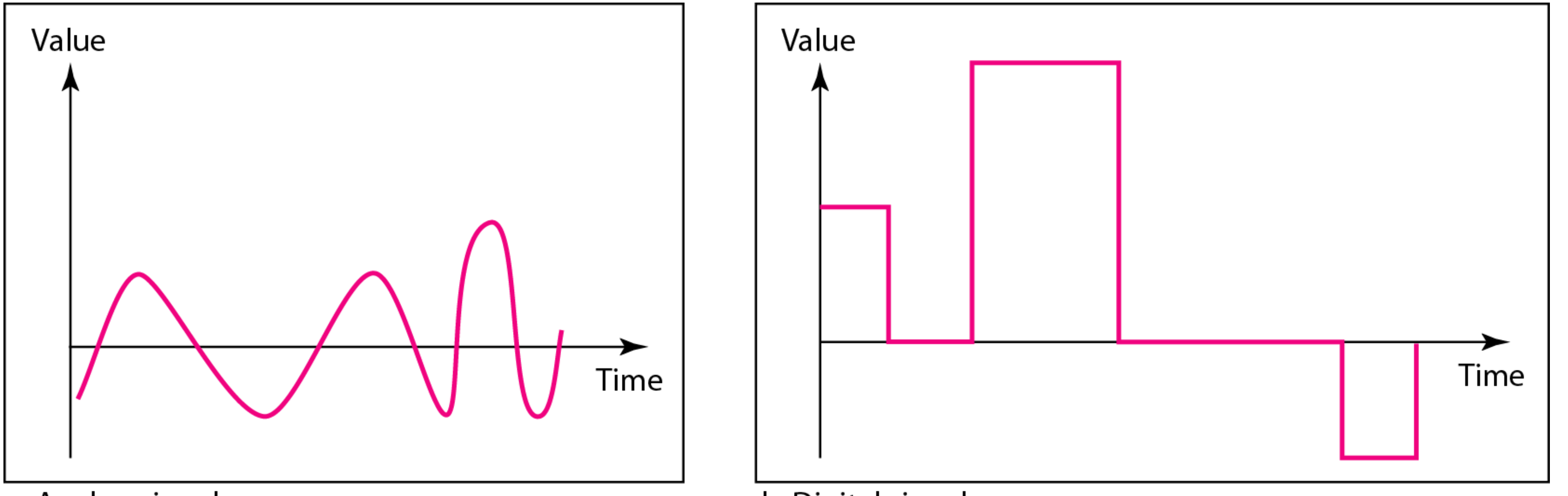
Data and Signals

- Data are entities that convey meaning (computer file, music on CD, results from a blood gas analysis machine)
- Signals are the electric or electromagnetic encoding of data (telephone conversation, we page download)
- Computer networks and data/voice communication systems transmit signals
- Data and signals can be analog or digital – this can be misleading – let’s talk more about this

Analog vs. Digital Signals

- Signals can be interpreted as either analog or digital
- In reality, all signals are analog
- Analog signals are continuous, non-discrete
- Digital signals are non-continuous, discrete
- Digital signals lend themselves more nicely to noise reduction techniques

Figure 3.1 *Comparison of analog and digital signals*



a. Analog signal

b. Digital signal

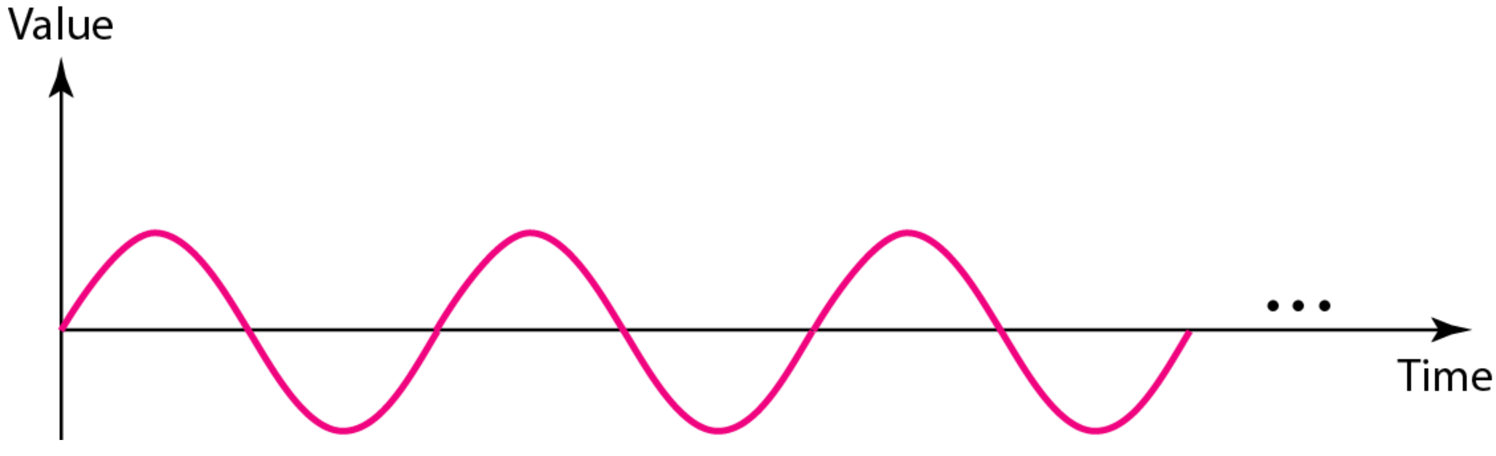
Time domain concepts

- Continuous signal
  - Infinite number of points at any given time
- Discrete signal
  - Finite number of points at any given time; maintains a constant level then changes to another constant level
- Periodic signal
  - Pattern repeated over time
- Aperiodic (non-periodic) signal
  - Pattern not repeated over time

Time domain concepts

- In data communications, we commonly use periodic analog signals and nonperiodic digital signals.
- Periodic analog signals can be classified as simple or composite.
- A simple periodic analog signal, a sine wave, cannot be decomposed into simpler signals.
- A composite periodic analog signal is composed of multiple sine waves.

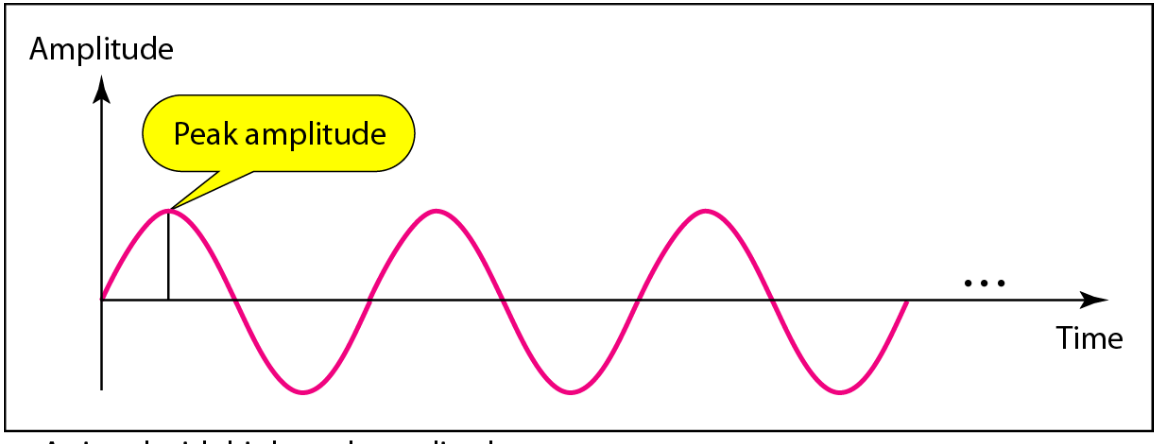
Figure 3.2 *A sine wave*



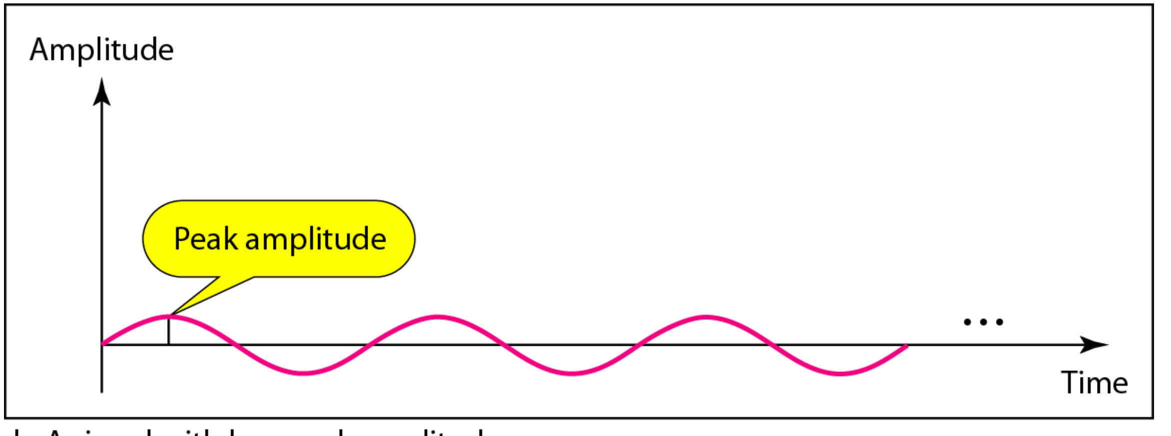
Signal Properties

- All signals are composed of three properties:
  - Amplitude
  - Frequency
  - Phase

Figure 3.3 ***Two signals with the same phase and frequency, but different amplitudes***



a. A signal with high peak amplitude



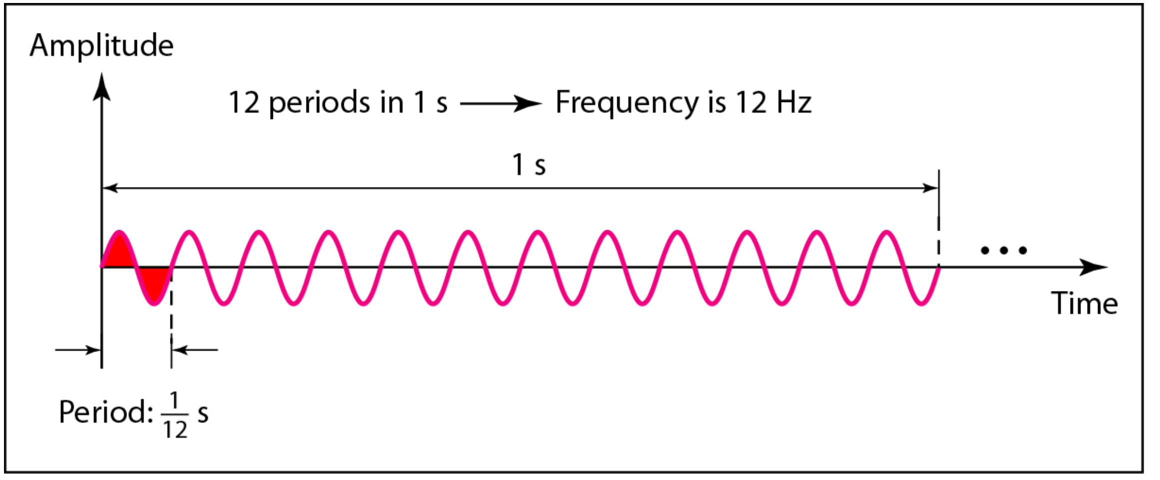
b. A signal with low peak amplitude

Note

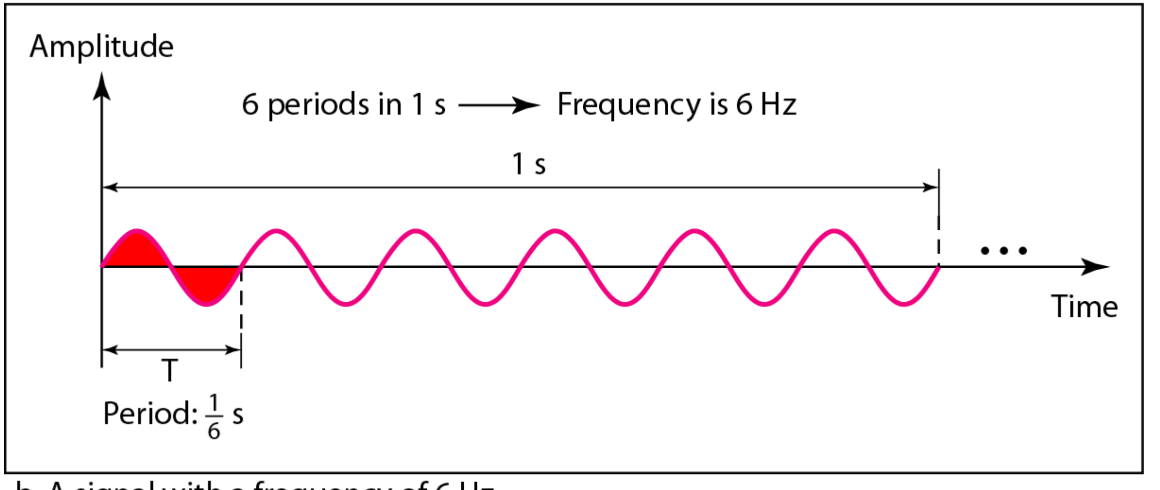
Frequency and period are the inverse of each other.

$$f = \frac{1}{T} \quad \text{and} \quad T = \frac{1}{f}$$

Figure 3.4 *Two signals with the same amplitude and phase, but different frequencies*



a. A signal with a frequency of 12 Hz



b. A signal with a frequency of 6 Hz

Table 3.1 ***Units of period and frequency***

Unit	*Equivalent*	*Unit*	*Equivalent*
Seconds (s)	1 s	Hertz (Hz)	1 Hz
Milliseconds (ms)	10^{-3} s	Kilohertz (kHz)	10^{3} Hz
Microseconds (μs)	10^{-6} s	Megahertz (MHz)	10^{6} Hz
Nanoseconds (ns)	10^{-9} s	Gigahertz (GHz)	10^{9} Hz
Picoseconds (ps)	10^{-12} s	Terahertz (THz)	10^{12} Hz

Example 3.3

The power we use at home has a frequency of 60 Hz. The period of this sine wave can be determined as follows:

$$T = \frac{1}{f} = \frac{1}{60} = 0.0166 \text{ s} = 0.0166 \times 10^3 \text{ ms} = 16.6 \text{ ms}$$

Example 3.5

The period of a signal is 100 ms. What is its frequency in kilohertz?

Solution

First we change 100 ms to seconds, and then we calculate the frequency from the period (1 Hz = 10^{-3} kHz).

$$100 \text{ ms} = 100 \times 10^{-3} \text{ s} = 10^{-1} \text{ s}$$

$$f = \frac{1}{T} = \frac{1}{10^{-1}} \text{ Hz} = 10 \text{ Hz} = 10 \times 10^{-3} \text{ kHz} = 10^{-2} \text{ kHz}$$

Note

Frequency is the rate of change with respect to time.

Change in a short span of time means high frequency.

Change over a long span of time means low frequency.

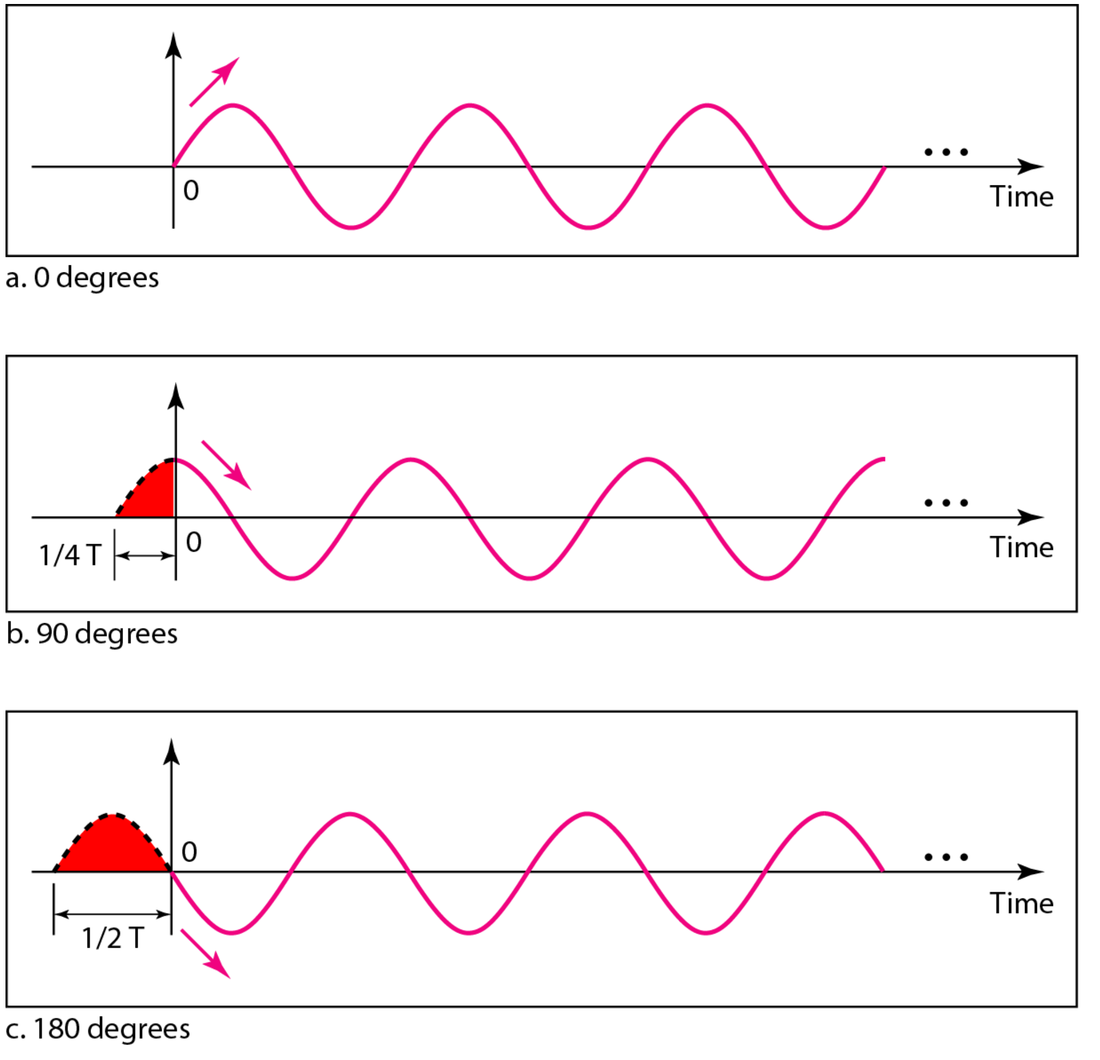
Note

If a signal does not change at all, its frequency is zero.
If a signal changes instantaneously, its frequency is infinite.

Note

Phase describes the position of the waveform relative to time 0.

Figure 3.5 *Three sine waves with the same amplitude and frequency, but different phases*



Example 3.6

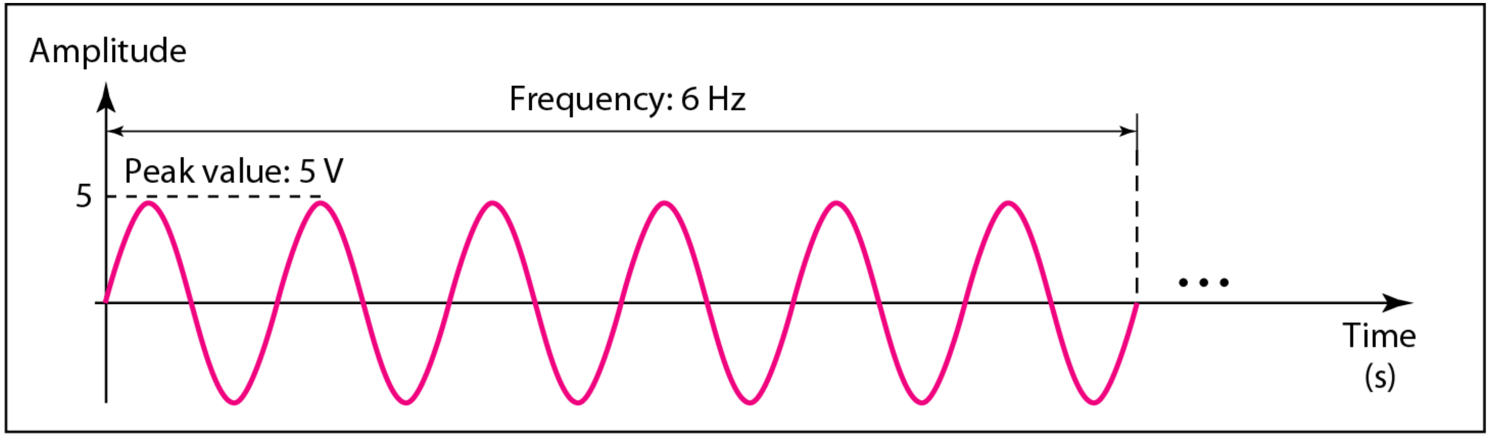
A sine wave is offset 1/6 cycle with respect to time 0. What is its phase in degrees and radians?

Solution

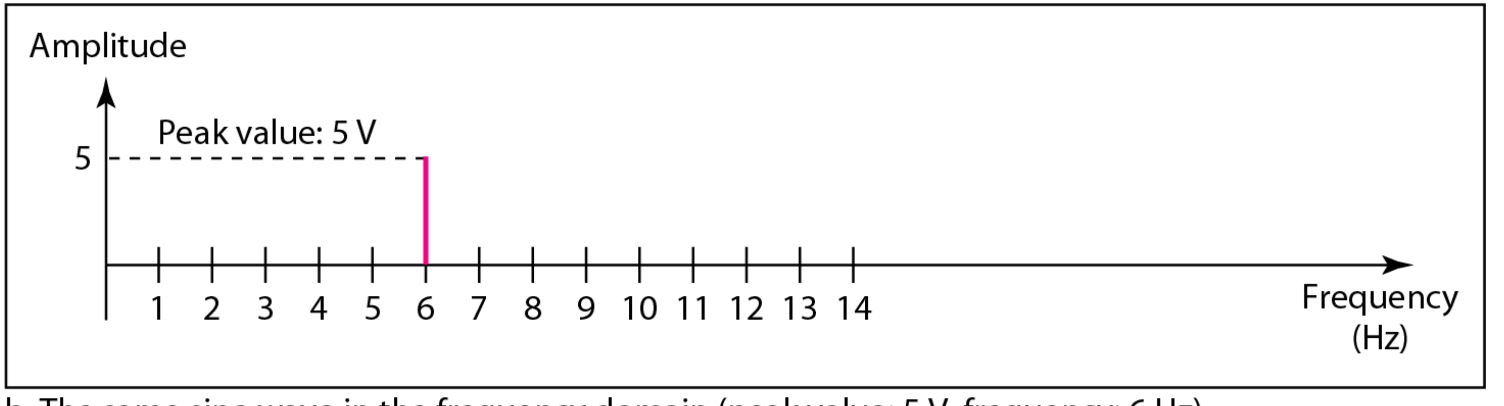
We know that 1 complete cycle is 360°. Therefore, 1/6 cycle is

$$\boxed{\frac{1}{6} \times 360 = 60^\circ = 60 \times \frac{2\pi}{360} \text{ rad} = \frac{\pi}{3} \text{ rad} = 1.046 \text{ rad}}$$

Figure 3.7 *The time-domain and frequency-domain plots of a sine wave*



a. A sine wave in the time domain (peak value: 5 V, frequency: 6 Hz)

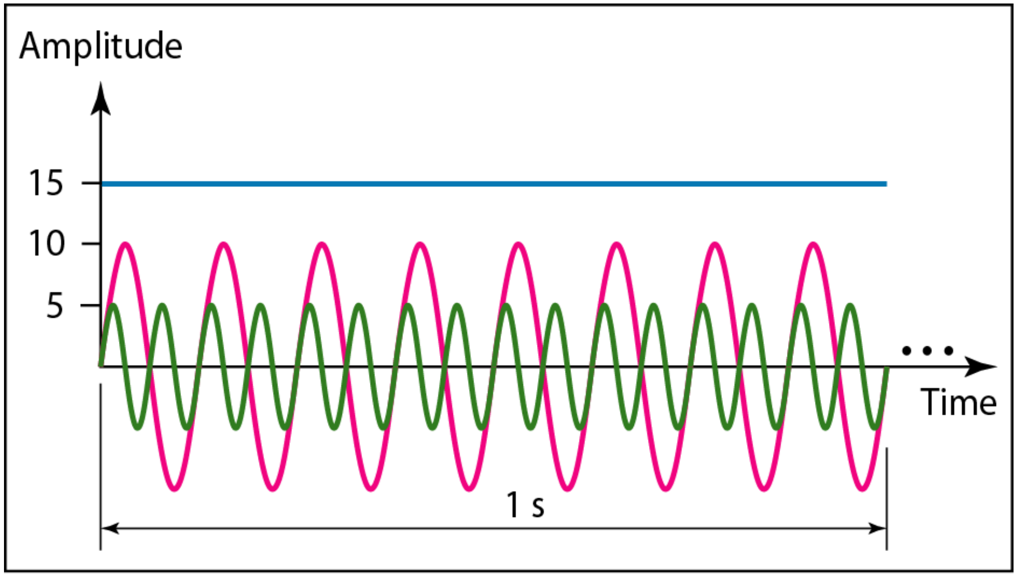


b. The same sine wave in the frequency domain (peak value: 5 V, frequency: 6 Hz)

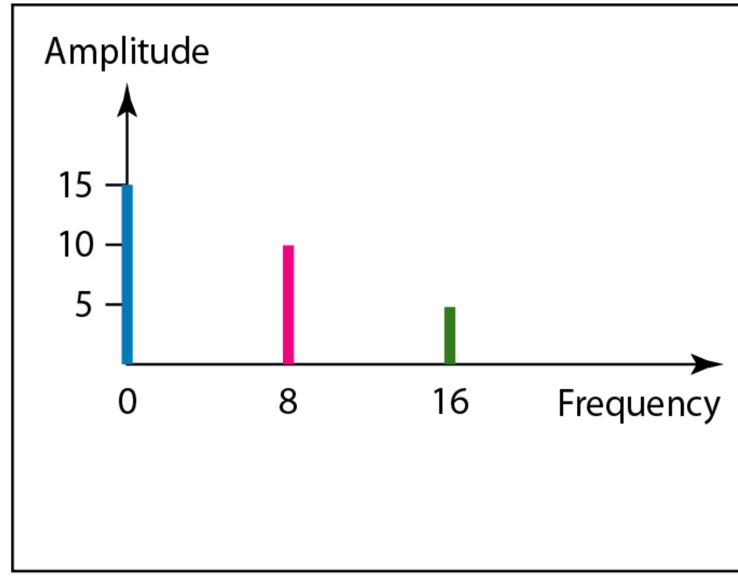
Example 3.7

The frequency domain is more compact and useful when we are dealing with more than one sine wave. For example, Figure 3.8 shows three sine waves, each with different amplitude and frequency. All can be represented by three spikes in the frequency domain.

Figure 3.8 *The time domain and frequency domain of three sine waves*

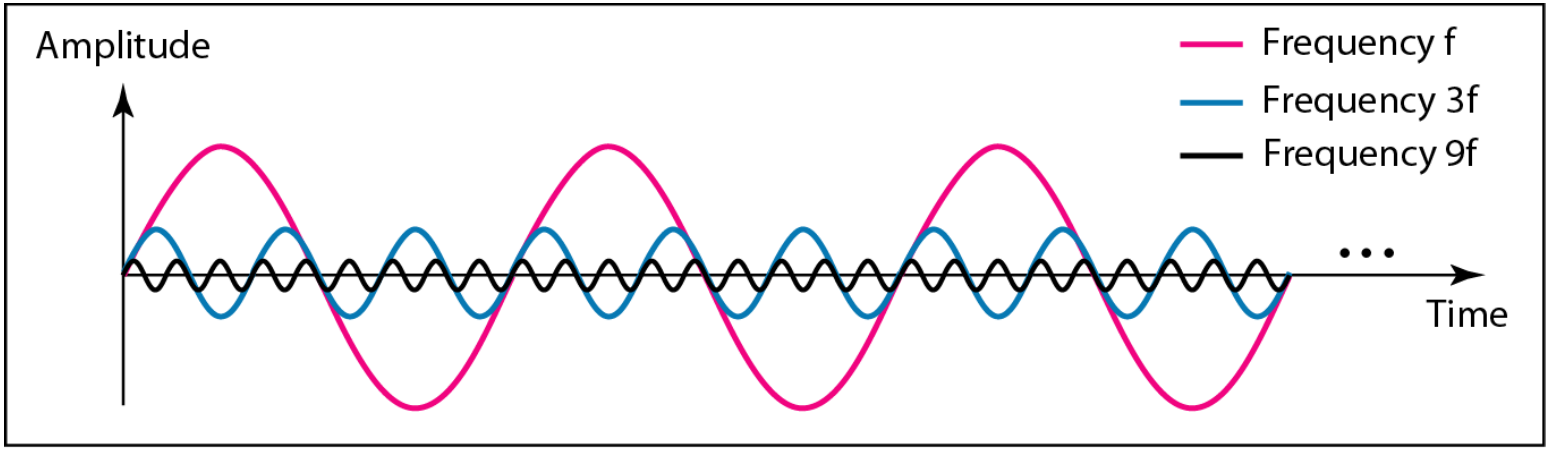


a. Time-domain representation of three sine waves with frequencies 0, 8, and 16

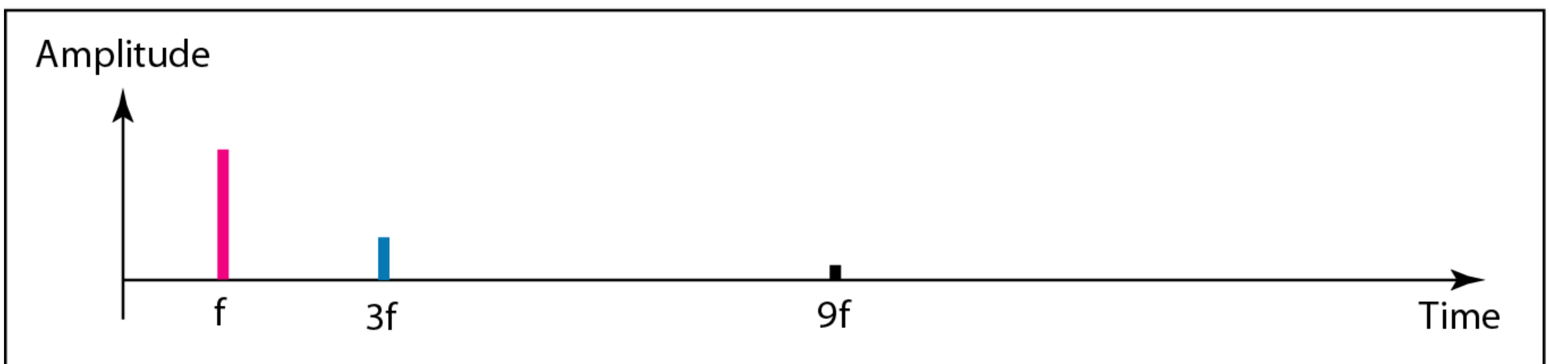


b. Frequency-domain representation of the same three signals

Figure 3.10 *Decomposition of a composite periodic signal in the time and frequency domains*



a. Time-domain decomposition of a composite signal

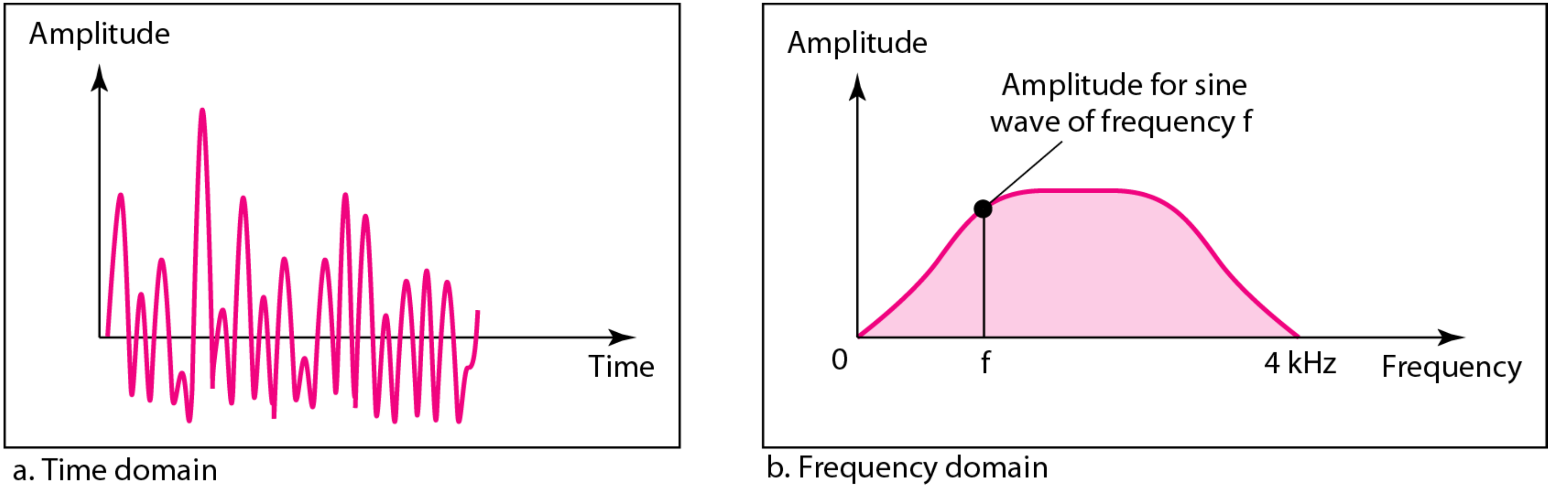


b. Frequency-domain decomposition of the composite signal

Note

A single-frequency sine wave is not useful in data communications; we need to send a composite signal, a signal made of many simple sine waves.

Figure 3.11 ***The time and frequency domains of a nonperiodic signal, such someone speaking into a microphone***

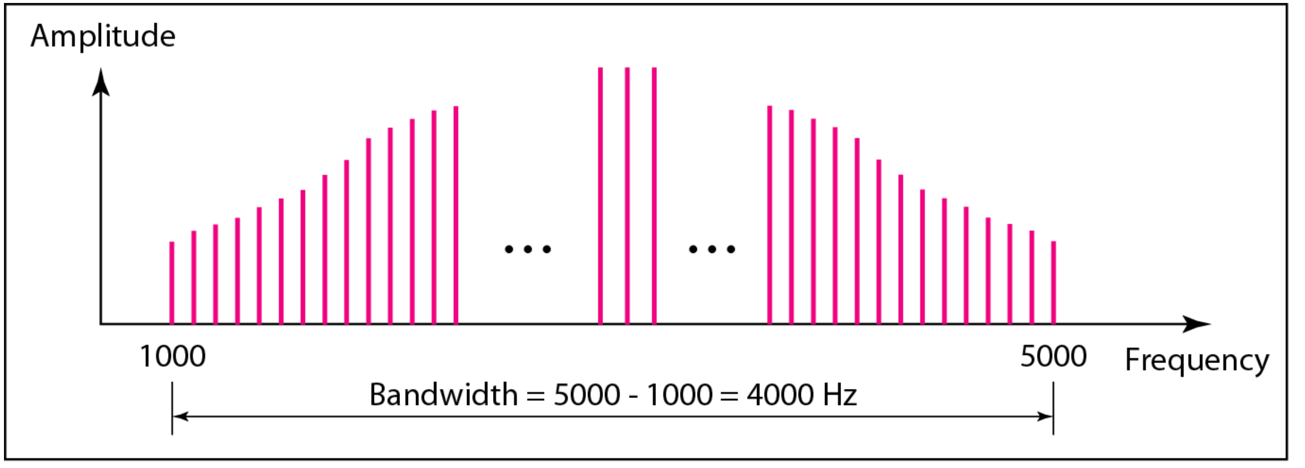


Note

The bandwidth of a composite signal is the difference between the highest and the lowest frequencies contained in that signal.

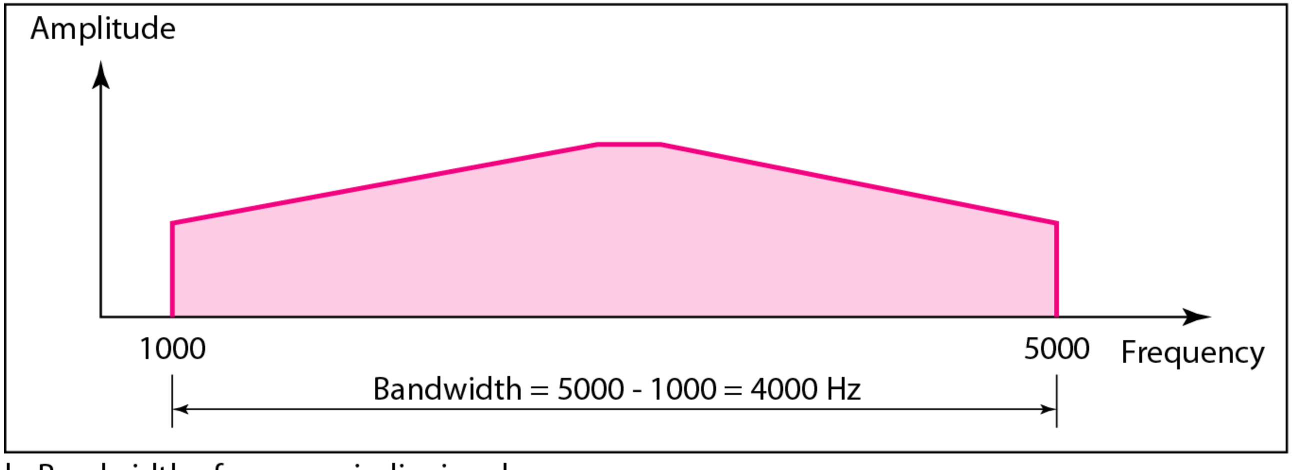
Figure 3.12 *The bandwidth of periodic and nonperiodic composite signals*

Note: each frequency is identifiable



a. Bandwidth of a periodic signal

Note: frequencies are all over the place



b. Bandwidth of a nonperiodic signal

Example 3.10

If a periodic signal is decomposed into five sine waves with frequencies of 100, 300, 500, 700, and 900 Hz, what is its bandwidth? Draw the spectrum (range of frequencies), assuming all components have a maximum amplitude of 10 V.

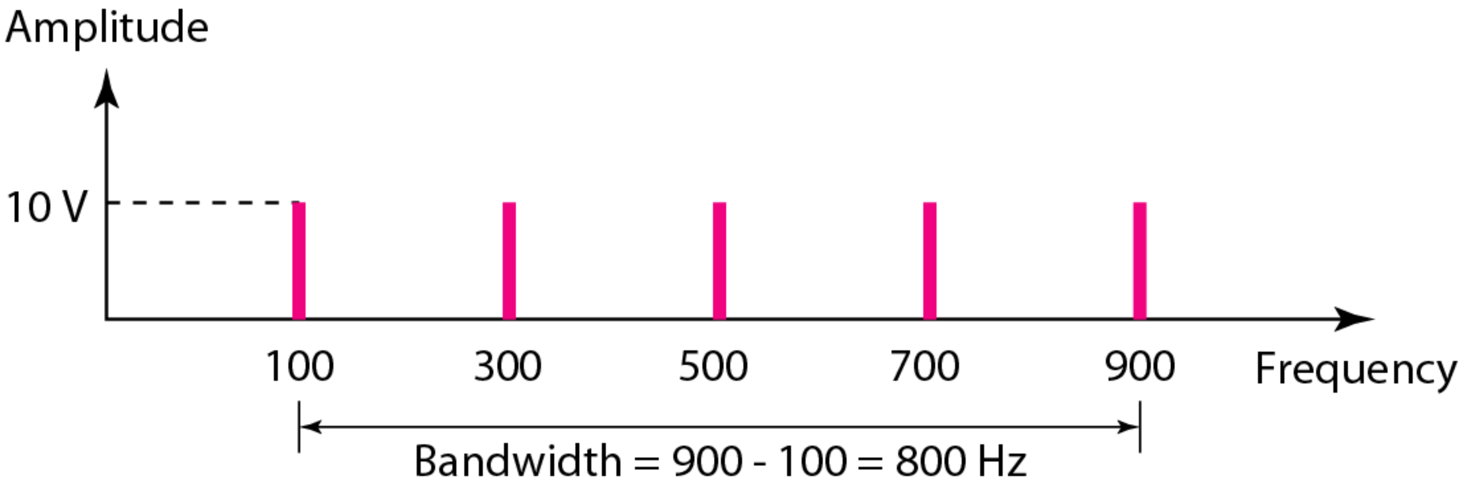
Solution

Let f_h be the highest frequency, f_l the lowest frequency, and B the bandwidth. Then

$$\boxed{B = f_h - f_l = 900 - 100 = 800 \text{ Hz}}$$

The spectrum has only five spikes, at 100, 300, 500, 700, and 900 Hz (see Figure 3.13).

Figure 3.13 *The bandwidth for Example 3.10*



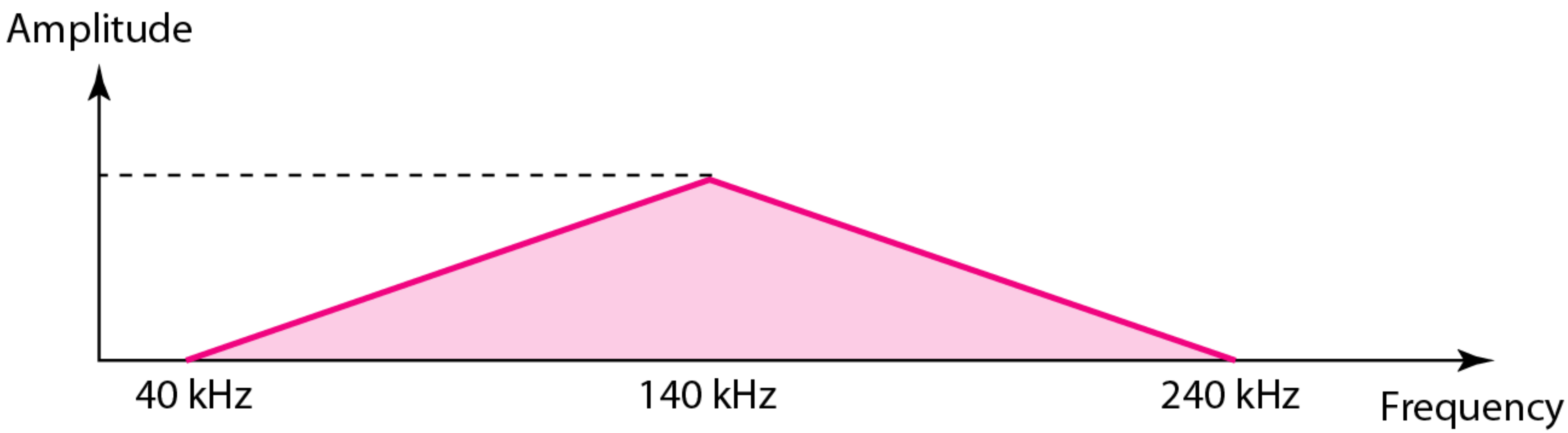
Example 3.12

A nonperiodic composite signal has a bandwidth of 200 kHz, with a middle frequency of 140 kHz and peak amplitude of 20 V. The two extreme frequencies have an amplitude of 0. Draw the frequency domain of the signal.

Solution

The lowest frequency must be at 40 kHz and the highest at 240 kHz. Figure 3.15 shows the frequency domain and the bandwidth.

Figure 3.15 *The bandwidth for Example 3.12*



1. Before data can be transmitted, they must be transformed to ________.

A. periodic signals

B. electromagnetic signals

C. aperiodic signals

D. low-frequency sine waves

Ans: B

2. A periodic signal completes one cycle in 0.001 s. What is the frequency?

A. 1 Hz

B. 100 Hz

C. 1 KHz

D. 1 MHz

Ans:C

3. In a frequency-domain plot, the horizontal axis measures the ________.

A. peak amplitude

B. frequency

C. phase

D. slope

Ans:B

4. In a time-domain plot, the horizontal axis is a measure of ________.

A. signal amplitude

B. frequency

C. phase

D. time

Ans:D

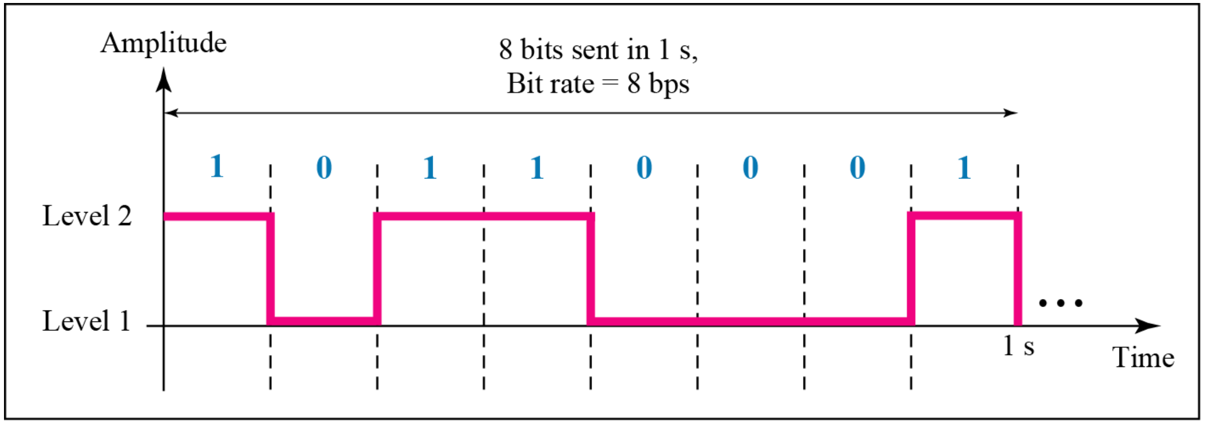
3-3 DIGITAL SIGNALS

In addition to being represented by an analog signal, information can also be represented by a digital signal. For example, a 1 can be encoded as a positive voltage and a 0 as zero voltage. A digital signal can have more than two levels. In this case, we can send more than 1 bit for each level.

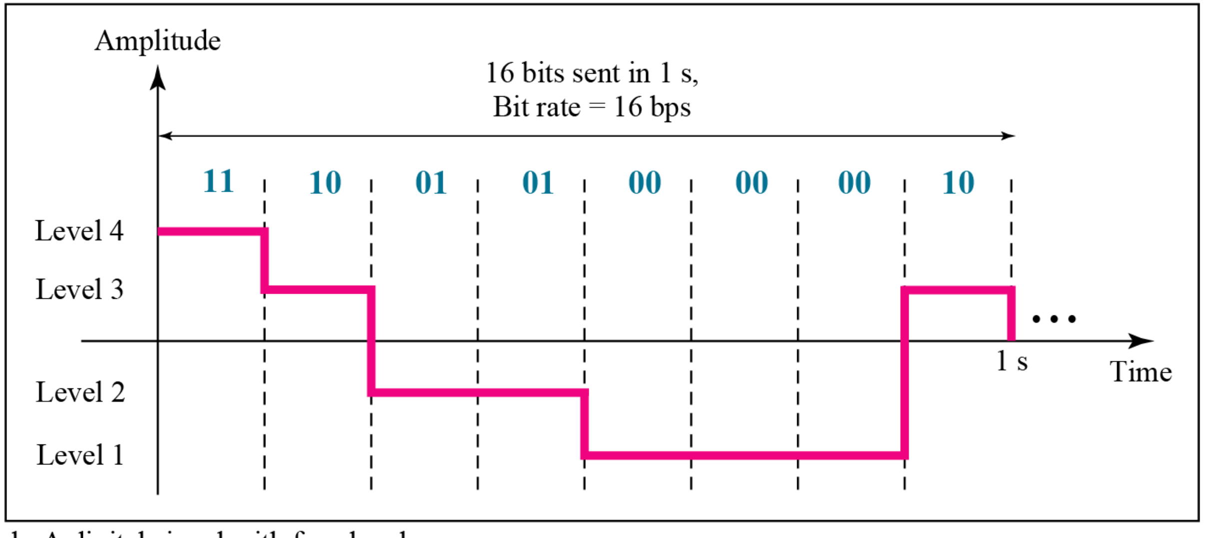
Topics discussed in this section:

Bit Rate
Bit Length
Digital Signal as a Composite Analog Signal
Application Layer

Figure 3.16 *Two digital signals: one with two signal levels and the other with four signal levels*



a. A digital signal with two levels



b. A digital signal with four levels

Example 3.16

A digital signal has eight levels. How many bits are needed per level? We calculate the number of bits from the formula

$$\boxed{\text{Number of bits per level} = \log_2 8 = 3}$$

Each signal level is represented by 3 bits.

Note

In baseband transmission, the required bandwidth is proportional to the bit rate; if we need to send bits faster, we need more bandwidth.

3-4 TRANSMISSION IMPAIRMENT

Signals travel through transmission media, which are not perfect. The imperfection causes signal impairment. This means that the signal at the beginning of the medium is not the same as the signal at the end of the medium. What is sent is not what is received. Three causes of impairment are attenuation, distortion, and noise.

Topics discussed in this section:

Attenuation
Distortion
Noise

TRANSMISSION IMPAIRMENT

Signals travel through transmission media, which are not perfect. This means that the signal at the beginning of the medium is not the same as the signal at the end of the medium.

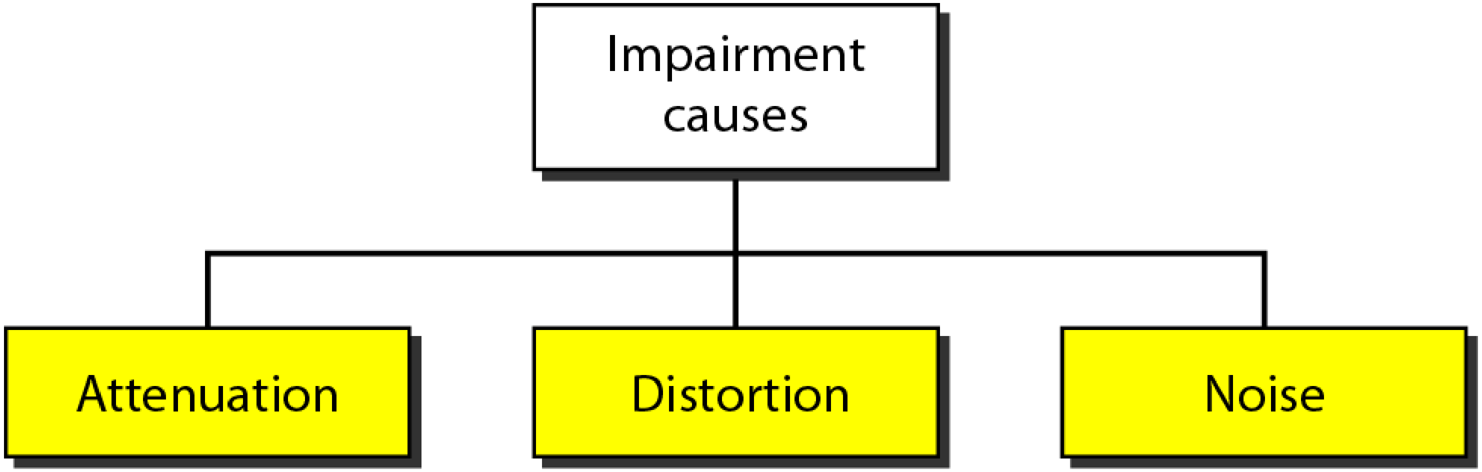
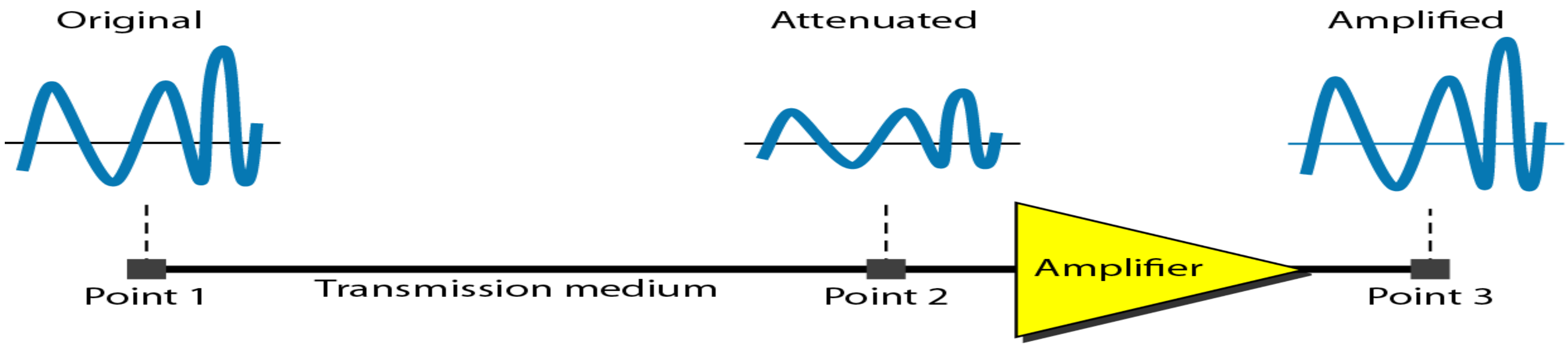


Figure : ***Causes of impairment***

Attenuation ***means a loss of energy. When a signal, simple or composite, travels through a medium, it loses some of its energy in overcoming the resistance of the medium.***



Example

Figure: **Attenuation**

Suppose a signal travels through a transmission medium and its power is reduced to one-half. This means that P2 is (1/2)P1. In this case, the attenuation (loss of power) can be calculated as:

$$\boxed{10 \log_{10} \frac{P_2}{P_1} = 10 \log_{10} \frac{0.5 P_1}{P_1} = 10 \log_{10} 0.5 = 10(-0.3) = -3 \text{ dB}}$$

A loss of 3 dB (–3 dB) is equivalent to losing one-half the power.

Distortion ***means that the signal changes its form or shape. Distortion can occur in a composite signal made of different frequencies.***
signal components at the receiver have phases different from what they had at the sender. The shape of the composite signal is therefore not the same. Figure shows the effect of distortion on a composite signal.

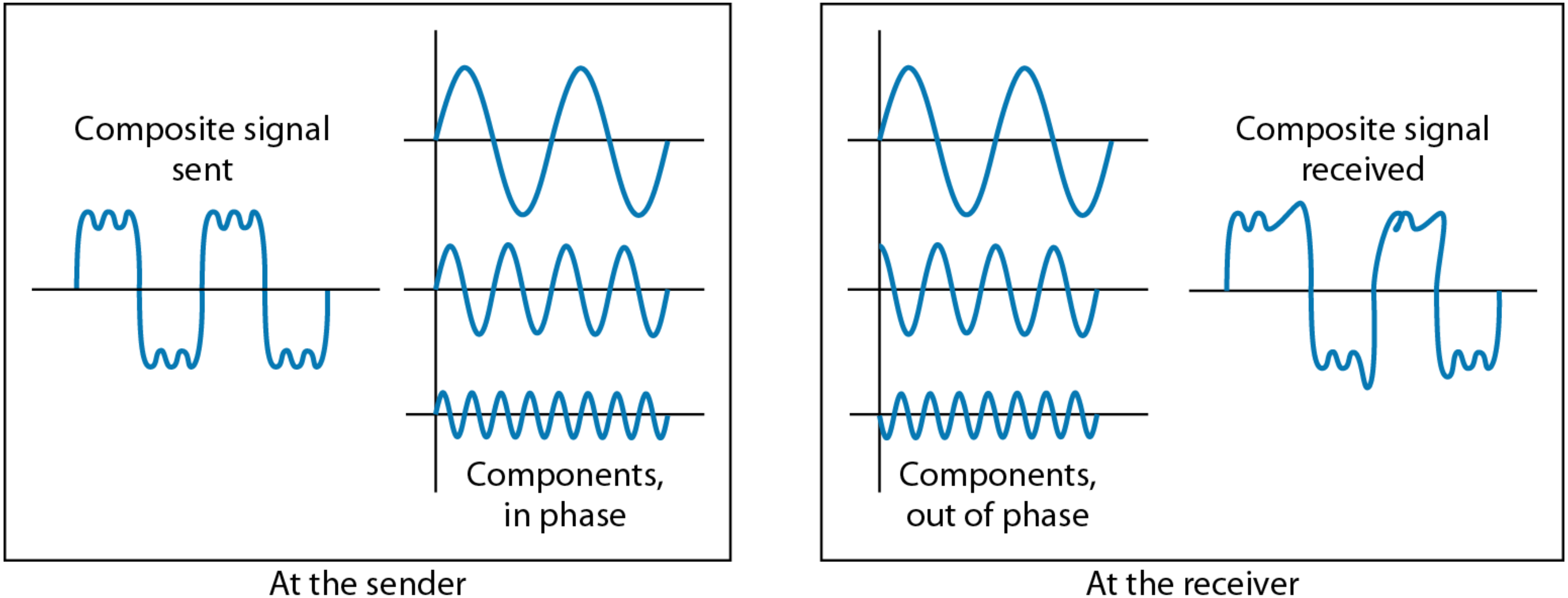


Figure: **Distortion**

Noise ***is another cause of impairment. Several types of noise, such as thermal noise, induced noise, crosstalk, and impulse noise, may corrupt the signal. Figure shows the effect of noise on a signal.***

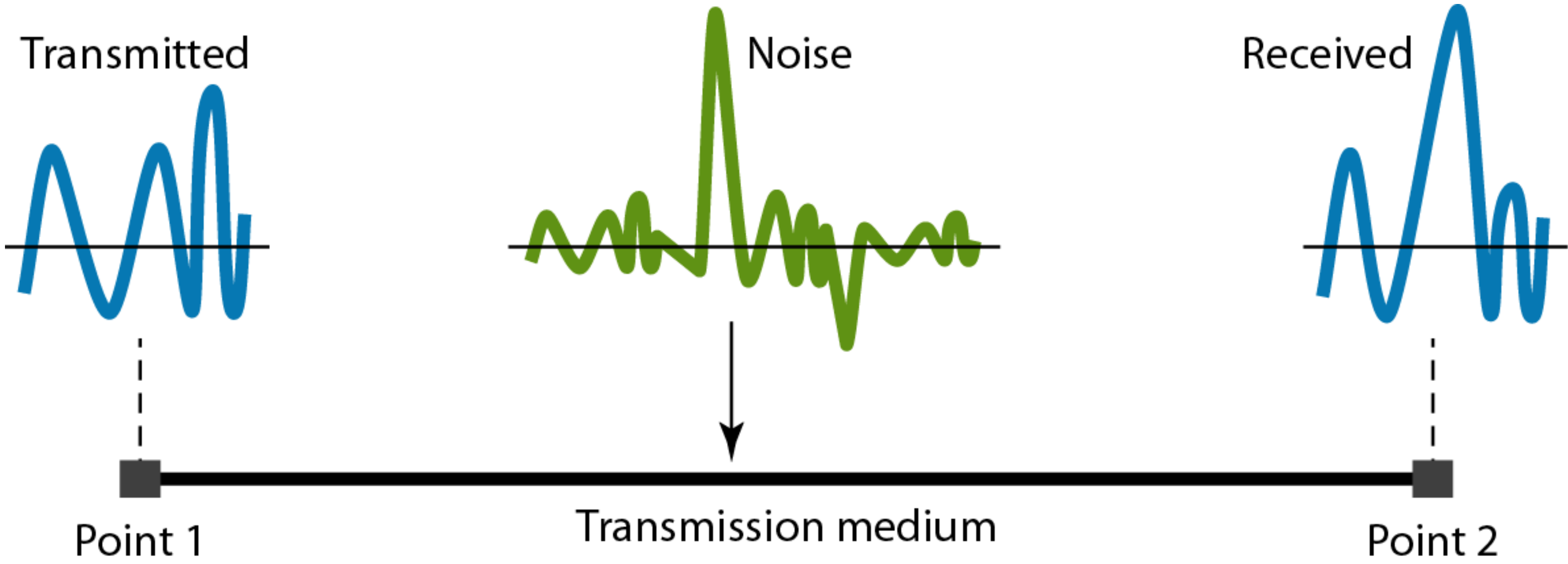


Figure : ***Noise***

Channel Capacity

- Data rate
  - In bits per second
  - Rate at which data can be communicated
- Bandwidth
  - In cycles per second of Hertz
  - Constrained by transmitter and medium
- Baud rate
  - Frequency with which the components change

5. If the bandwidth of a signal is 5 KHz and the lowest frequency is 52 KHz, what is the highest frequency?

A. 5 KHz

B. 10 KHz

C. 47 KHz

D. 57 KHz

Ans:D

6. What is the bandwidth of a signal that ranges from 1 MHz to 4 MHz?

A. 4 MHz

B. 1 KHz

C. 3 MHz

D. none of the above

Ans:C

7. As frequency increases, the period __________.

A. decreases

B. increases

C. remains the same

D. doubles

Ans:A

9. A sine wave is __________.

A. periodic and continuous

B. aperiodic and continuous

C. periodic and discrete

D. aperiodic and discrete

Ans:A